

Tempestology Part 1 with Matt Lanza & Dr. Kim Wood

Ologies Podcast

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Oh hey, it's your ex-boyfriend who pretended that he didn't know how to wash the cast iron right, so he just didn't, Alie Ward. This one is a smooth sail through rough weather, friends. We get hurricanes. You get hurricanes. Maybe you have typhoons or cyclones. Let's hear about the differences, let's hear how fast they go, how many we might have, and some real historical whoppers, and more, with not one, but two tempestologists.

So, tempestology, it's a real word. It describes the study of cyclones, and hurricanes, and other extreme weather events. And the word 'tempest,' I just found this out, it comes from the Latin root for temporal, meaning time, which morphed into seasons, which morphed into weather and storms, which is apt because now is definitely the time on Earth to hear about hurricanes... Whooooo Nellie, is it.

So, the first ologist we talked to studied physics and geophysics at Oregon State University, and then got a Master's and a PhD in atmospheric science at the University of Arizona. They're now an associate professor in the Department of Hydrology and Atmospheric Sciences at the University of Arizona, and they teach about tropical meteorology and their research of these giant storm systems. So, a shoo-in. And because they both came so highly recommended, and I couldn't choose, we also have a Rutgers-trained forecast meteorologist in Houston, and a writer and editor for Houston's highly-respected meteorological outlet, Space City Weather. This guest is a co-founder of the website, The Eyewall, too, which keeps an eye on developing storms, and it tells you, without a lot of hype, just straight-up what to expect.

So, both of these experts, they're so passionate about hurricanes and relaying info to the public. And I chatted with them both, and I heard so many stories that stuck with me. I got so much valuable info that we had to make this a two-parter.

But before we dive in, thank you to patrons of the show, who make *Ologies* possible. You can join for a dollar or more a month and send in questions to the ologists via Patreon.com/Ologies. Thank you to everyone out there in merch, from OlogiesMerch.com. Thank you to everyone who leaves a review for me to read, which helps keep us up in the charts. I do read all of them, like some weird old-timey man with a candle and a monocle, such as this one from the elegantly named HotdogHarriet, who wrote:

What do you want to know? Anything. If you want to know about anything, this is the pod for you. When my family or I have a question about something, one of the things we do now is check to see if there's an episode of Ologies to answer that question or dig deeper.

Hotdog Harriet, thank you for digging into my archive. If you too want to know which 400-plus episodes we have just waiting for you, go to Ologies.com. It's all neatly sorted into topic. Also, if you do not like swear words, we have a spinoff show called *Smologies*, and you can subscribe wherever you get your podcasts or at the link in the show notes.

Okay, Tempestology. Batten your hatches for a Part One all about cyclones, and hurricanes, and typhoons, oh my! Where hurricanes come from, what your news meteorologist even means by high- and low-pressure systems, why hurricanes have eyes, the bizarre history of the category system, the also bizarre history behind the naming system, if lady names mean more deaths, how we figure out the GPS of a big old storm a brewin'. And next week, we're going to be back to sort out the best

ways to prepare for storms like this, what is happening with climate change, the latest on this year's forecast, the latest on the funding drama, if you should trust a Waffle House for your news, and literally, what is on the horizon as hurricane season rears at the gate of its busiest time with two meteorologists and hurricane experts and tempestologists, Dr. Kim Wood and Matt Lanza.

Kim: I'm Kim Wood. I'm an associate professor at the University of Arizona. I specialize in atmospheric dynamics, and my pronouns are they/them.

Alie: Now, hurricanes in Arizona. Are there any?

Kim: Technically, yes, but the leftovers?

Alie: Oh! Okay.

Kim: So, because the storms don't like moving over land, they tend to fall apart when that happens, you're not going to still have a coherent "hurricane" by the time it gets to Arizona. But the leftover moisture plume will come in and potentially drop quite a bit of rain. In 1992, Hurricane, then Tropical Storm Lester, came right over Tucson! But it coincided with a slightly bigger news story of Hurricane Andrew hitting Florida as a Category 5. So, it's a bit less talked about than the devastating Andrew.

Aside: And yes, that was late August 1992. And I will *happily* deal with any ideo of any March, but when it comes to late August, you're going to catch me away from a coast. But from the Category 5, Andrew, to Matt.

Matt: I am Matt Lanza, he/him.

Alie: And a meteorologist, yes?

Matt: I am a meteorologist, yes.

Alie: And, I found this out, a tempestologist.

Matt: Oooh!

Alie: It's a thing. Did you know that? That tempestology is hurricanes?

Matt: No, I did not know that.

Alie: And it's the study of cyclones.

Matt: Okay!

Alie: Isn't that cool?

Matt: Well, I feel even cooler now. *[laughs]*

Alie: And, you know, we're here in the US and we call them hurricanes. But I understand they are cyclones in other places, tropical cyclones.

Kim: So, tropical cyclone is what we'd call the generic term for all of these systems. They then get more specific names depending on where they are. So, if they're in the North Atlantic or Eastern North Pacific, they're hurricanes once they reach 74 miles an hour.

Aside: So, the North Atlantic, those are the hurricanes that hit the Caribbean, the east coast of North America and the Gulf Coast. Now, the eastern North Pacific, that took me two maps to understand, but it is the area to the west of Mexico, like, near the Baja Peninsula. The Eastern North Pacific, this region, it is a very overlooked basin. People, I live in LA, and I just found out about it now. But yes, those storms are all called hurricanes.

Kim: If they're in the western North Pacific, they're typhoons. And then in the southern hemisphere, they tend to use the broader term cyclone. Sometimes they'll say, "severe cyclonic storm." I think that's in the Indian Ocean. But broadly speaking, if it's a tropical depression, tropical storm, hurricane, typhoon, they're all categories of tropical cyclone.

Alie: And then what about a tornado? Is a tornado a cyclone of any kind?

Kim: It is a cyclonic system, but it is very, very, very tiny by comparison. A hurricane can produce tornadoes. [*"Yikes."*] Tornadoes cannot produce hurricanes.

Alie: Okay. And do hurricanes have more moisture in them than tornadoes? Or what makes a hurricane a hurricane?

Kim: Oh, uhhh... Do you have a semester? [*both laugh*] What makes a hurricane a hurricane is they're a type of cyclone, and they have a low pressure—so they're lower in pressure at the center compared to on the outside—and they have something called a warm core.

Aside: Okay, this is fun. And I did not know this, but a low-pressure system means that this column of air is rising, kind of like getting sucked up into the sky in a vacuum. And low-pressure systems are rougher air, clouds, precipitation, and all that unstable air is what can cause turbulence in a plane, and what drives hurricanes. Now, a high-pressure system, by contrast, is like a hairdryer blowing down toward the Earth. High-pressure systems tend to bring calmer weather. Some meteorologists have this casual shorthand that high pressure is happy weather, and low pressure is lousy weather, but that depends on your preference for rain. Personally, I'd flip that.

But yeah, hurricanes, low pressure. And Kim describes it first by what it doesn't do. Like, let's say someone hosed you off and it just felt delightful. That's because the water on you is borrowing your body heat.

Kim: You know, if it's really hot out and you get some water on you, you're like, "Ahhhh." You're finally cooling off. Well, that's evaporative cooling, so energy is being taken from your body to convert that liquid water into water vapor. Well, when it goes the other way, when it condenses from water vapor into liquid water, that heat is released into the air around it, and that helps drive those billowing clouds that you see. A hurricane is also powered by that kind of, what we call, moist convection and that latent heat release. And so, that latent heat release contributes to the center of the cyclone being warmer than the air around it, and so that's why we call it a warm core. Physics dictates that when you have a warm core system, the strongest winds are closest to the surface, which is where we live, which is not great. But that's also how we measure how strong a storm is. What are those wind speeds closest to the surface, closest to the center?

Alie: And that's closest to the surface of the Earth. So, closer to where the sky intersects land, and then we're the houses and people right on that meridian.

Kim: Yep, yep. It gets slowed down a little bit because of friction. But that's why if you see reports from a hurricane hunter flying through a storm, and it's like, "Oh, we measured winds of 100 knots," but at the surface, it was like 85. It's not because the storm's, "weaker" or something. It's because friction slows down winds, and the closer you are to the surface, the more friction is going to slow those winds down.

Alie: What's up with those hurricane plane hunters? [*airplane engine rumbling "Damn."*] Because I've seen videos of things that look like a nightmare. [*"Oh god."*] Who goes in those? Why? What did we learn? How do they not die? How do they not fall out of the sky?

Kim: [*laughs*] Well, they might have fallen out of the sky back in the 1940s.

Aside: So, Kim is not kidding. Between 1945 and 1974, there have been at least six crashes from hurricane hunter missions. Kim explained that because those low-pressure systems are what they're flying into, flying too low into it means not enough air to fly on, or reduced lift, and a lot of turbulence. Nowadays, though...

Kim: But these hurricane hunters, if you hear about the Air Force or NOAA, they're flying at about three kilometers above the Earth's surface, so they're low enough that they're getting really into the heart of the storm as far as vertically, where the strong winds are. But they're high enough that they're not going to run into the ground. So, they are well-equipped with radar and other instruments, so they can navigate. They've got expert pilots on board, they have expert scientists on board, and all sorts of folks with experience in running these instruments, dropping dropsondes, which are the opposite of what we put on weather balloons; they get a little parachute, they drop out of the plane, and then they take measurements as they fall down to the ocean surface below. But the folks on there tend to be a range of veterans with lots of—when I say veterans, like, having many years of piloting experience—but also meteorologists who have spent a lot of time in the classroom and beyond studying how hurricanes work.

Aside: That is so amazing. I would never want to do that. What about Kim?

Kim: Back in 2010, I got to do that on the NASA DC-8. [*Alie gasps and groans*] So, we were way up beyond 30,000 feet, so way above the level that you would fly in a typical hurricane hunter, but I flew through Hurricane Karl in 2010 as it was rapidly intensifying, and yeah, it's bumpy. I was very happy to have a four-point harness kind of seatbelt.

Alie: What is it like? What happens to that turbulence? Does it reduce the closer you get to the eye of the storm? What's the mood like on one of those?

Kim: [*laughs*] Well, I would say the mood's pretty excited because we're flying through a hurricane. Like, we're scientists, but we're also nerds, and so we're just excited to learn all these things. They'll often put instruments on these planes so that we can learn more about how they operate before we try to put them in space on a satellite, because getting them into space could be hundreds of millions of dollars. But anytime you go through an area with a strong updraft—so that's what's driving upward those thunderstorm-type clouds—you can get some real, real interesting experiences, like an updraft-downdraft couplet where you're very briefly weightless.

Alie: Oh god. What a bucket list item to check off for a weather nerd.

Kim: Oh, yeah.

Alie: You know, like, that's just as cool as it gets. But have you always been kind of a weather person or a space person? Or when did you become like, atmos-Kim?

Kim: [*laughs*] So, one thing I want to preface this with is there's kind of this stereotype that weather weenies are into weather from, like, as soon as they could walk. And I want to encourage anyone who might hear this, if you're interested in weather and it starts when you're, like, 50 years old, we welcome you too. You don't have to be a weather weenie when you're a kid to be a weather weenie as an adult.

That said, I started getting interested in clouds when I was, like, 8 years old. But then, when I was a teenager, I went to a local air show, and they had a hurricane hunter crew there as, like, one of the exhibits. I wandered the plane and talked to the flight meteorologist. And I was just like, "This is so cool! I'm going to go study physics and then atmospheric science, and I guess I'm going to be a professor and teach other people about this too." [*"Myself included."*]

Aside: Okay, let's meet Kim's friend and colleague, Matt Lanza, who is a Houston-based meteorologist who quit a career in broadcast meteorology to help provide residents in storm-prone areas with some “hype-free forecasting,” as he calls it. He runs Space City Weather alongside fellow meteorologist Eric Berger. So, how valuable are they to public weather knowledge? They received the Houstonians of the Year Award in 2024, that's how much.

Alie: We are recording this July 25, 2025, and around the corner, August, is when, at least in North America, things start picking up, right?

Matt: Yeah, exactly. The peak of hurricane season is August and September. Usually, the last couple of weeks of August and the first couple of weeks of September are usually the most frenetic.

Alie: You are on the Gulf Coast, which is, like, woof! Right?

Matt: I grew up in Southern New Jersey, just outside of Atlantic City. I remember when I was 3 years old, we had Hurricane Gloria come up the coast, this was 1985, and we had to evacuate my grandmother, who lived in Atlantic City on a barrier island, to bring her onto the mainland. And just something about that moment, I think, triggered, you know, my interest in meteorology and hurricanes, and I guess it's kind of fate that I'm here in Houston now.

Alie: And now, Houston gets walloped as well, yes?

Matt: Yes, regularly, by everything. Not just hurricanes, but all sorts of weather, and hurricanes just are kind of the top of the pyramid for us in terms of threats.

Alie: Have you been in one?

Matt: Yes. Last year, Hurricane Beryl, 2024, was, I'll use scare quotes, “Only a Category 1,” but it was a nasty Category 1 storm; did a lot of damage, knocked out a lot of power. It was interesting to go through because I had actually not been in an actual hurricane— I've been in some tropical storms before, but going through a hurricane was kind of a whole other experience, not necessarily one I want to go through again.

Alie: Yeah. What was the most surprising from being an expert in this and studying them versus being in it and seeing it right outside your window? Did you feel any different, or did it feel like, “Oh yep, this is exactly what I expected.”?

Matt: Yeah, it felt different. When you think of a hurricane, you think it's just a constant, ferocious wind that's continuously coming, and it wasn't that. There were gaps where it was calm, and then all of a sudden, you would just get these nasty wind gusts, and that's what knocked down tree branches and limbs and things like that. You kind of know that as a meteorologist, that that's kind of how hurricanes go, but to actually experience it gave me a newfound appreciation for what it's like to just kind of sit through it. It's a grind, right? You're just kind of, as soon as you think it's done, you know, another round of gusts come and you're like, “Oh, here we go again.” It was fascinating in that regard.

Aside: Okay. So, we usually do patron questions after the break; you might be used to that. But kind of like a windsock and a tempest, we are all over the place format-wise for this one. I have confidence you can handle it. But let's keep on the theme of things being fascinating.

Alie: In terms of formation, Heather Crane wanted to know: I've always heard that hurricanes start as dust storms in Africa. How true is that?

Matt: It's kind of actually fascinating. You know, we always think about, yeah, the storms come off Africa, they go across the ocean, they end up hitting land, and all this stuff. So, first off, with the dust, the Sahara Desert in Africa is obviously going to be a massive source of sand, dust, et cetera. And

during, usually the early part of hurricane season, June and July, typically what happens is everything flows from, like, northern Africa, across the Atlantic, into North America, into South America, all that. So, what happens is, you get all this dust from Africa that gets picked up from the Sahara and blown across the ocean. In places like Houston, Miami, New Orleans, during hurricane season, you'll see, frequently, just dusty skies. The sky turns a little bit of, like, a milky-gray color, and this is all Saharan dust that's been transported halfway around the world.

Alie: Wow! Oh my god!

Matt: Yeah, it's fascinating. And they deal with this a lot in Puerto Rico and in the Caribbean islands as well. What's really cool about it is it's a huge part of the planetary ecosystem. It ends up helping to enrich soil on this side of the world. So, like, they found evidence in South America that, I think in the Amazon, there's a lot of fertile land, and a lot of it ends up like almost being seeded by the dust that comes off of Africa to help make it more fertile.

Alie: What?!

Matt: Yeah! It's so cool. But what it also does is it inhibits hurricanes because hurricanes need moisture. They need moisture, and dry air is dust. Dust comes from the desert, desert is dry, that's dry air. Hurricanes don't like that. So, that explains a lot why the first part of hurricane season is usually typically quiet because we're often seeing, you know, all this dust come off Africa and basically limit the development of storms. So, that's a really cool feature.

As we get into August and September and things start to ramp up, you get these thunderstorm complexes that start to form in Africa, and they move all the way across the continent in a similar vein to kind of what we see in Central America during this time of year, parts of South America where you get, you know, thunderstorms blow up during daytime heating, you can almost time your clock to having thunderstorms and they move through and that's that.

Aside: And these complexes, or storms, form and they dissipate. When they travel over the ocean, they are called, very confusingly, waves... but waves of air. I don't know why, of all the words, they didn't think of another name, but I'm not the boss.

Now, quick note. Remember that we call these cyclonic storms "typhoons" when they're in the Northwest Pacific Ocean; those are the coasts of Japan, Korea, China, the Philippines, and that is known as Typhoon Alley, which sounds like it could be the absolute worst dive bar in San Diego or something. But activity in Typhoon Alley, that can depend on whether we have a warmer El Niño year or a colder La Niña. And remember that they are measured differently, but essentially, anything over what we would call a Category 4 is called a Super Typhoon in that Northwest Pacific basin.

Now, as for the Atlantic hurricanes, El Niño (or warmer weather) means potentially fewer large cyclonic storms. Now, for the Northwest Pacific basin, again, that's Typhoon Alley on the coast of Asia, El Niño means more and bigger typhoons. Now, sometimes, in North America, Matt told me, we may have a Midwest storm that just goes and takes a dip in the Gulf of Mexico, picks up water, and then becomes a bigger storm, and then, bam, a hurricane.

Also, let's not forget about storm systems off the coast of India and in the Arabian Sea and the Indian Ocean, which are just called cyclones. We don't call them typhoons, we don't call them hurricanes necessarily, we just call them cyclones. Now, 1970's Bohla Cyclone, which formed in the Bay of Bengal, which is east of the Indian Peninsula and west of Thailand, it made landfall in what's now known as Bangladesh. And the flooding into the Ganges River basin caused an estimated half a million deaths. It was the worst natural catastrophe in recorded history. And the fallout from it helped spark 1971's Bangladesh Liberation War, which led to the ethnic cleansing and genocide of

Bengalis, and up to three million deaths. Now, for more on genocide, including recent events in Gaza, we have a Genocidology episode with expert Dr. Dirk Moses, and it's linked in the show notes.

But on the topic of depression, let's back up a skosh.

Alie: But when it comes to hurricanes, for those of us who are lucky enough to never have had to hide from one, I think we hear, like, it starts as a Category 1 tropical storm, or tropical depression. Where's the spectrum of when it becomes a hurricane and then becomes not a hurricane, or a typhoon, or a cyclone?

Kim: Yeah. Absolutely great question. So, the spectrum of what constitutes a tropical cyclone encompasses pretty much everything you just listed. So, let's traverse sort of a typical life cycle of a storm that might brew in the Atlantic.

Over Africa, we have these systems that are called African easterly waves, which is a fancy name for globs of thunderstorms that are relatively organized. They spin up off of the instability that happens between the moist, tropical latitudes over equatorial Africa versus the hot, dry Saharan Desert. But the energy that feeds them when they're over Africa, well, it's a little different once they're over the ocean. If they can start feeding off the energy of the ocean and instability in the atmosphere over the ocean, then they can start to consolidate around a center and spin up into a cyclone. So, they're called cyclones because they're spinning around a particular point. At that stage, they've gone from a disturbance that we're keeping an eye on, to a tropical depression.

Aside: So, a tropical depression is a low-pressure system, but with much weaker winds, just like the most mellow kind of hurricane. Just think depression, think slow, like *[softly]* "Mehhh."

Kim: So, once they get that tropical depression label, they are considered tropical cyclones. On the Saffir-Simpson hurricane wind scale, we don't use that scale until they've hit 74 miles an hour, which is a Category 1. So, a tropical storm is 39 to 73 miles an hour, which sounds kind of arbitrary, but that's the range that we use. So, that means a 70-mile-an-hour tropical storm is only slightly weaker, so to speak, than a 75-mile-an-hour hurricane. But we start paying attention when it flips that category from a tropical storm to a hurricane. We also pay attention when it goes from Category 2 to Category 3, they are major hurricanes.

Aside: So, the Saffir-Simpson Hurricane Wind Scale, it comes on the scene in 1971 after a guy named Herbert Seymour Saffir, a civil engineer, was tasked with figuring out how to build low-income housing that wouldn't blow away in a storm. He's like, "Well, how strong a storm, people?" So, he teamed up with a meteorologist and hurricane hunter, went up in those planes, Robert Simpson, to come up with a scale. Now, when you hear about the Saffir-Simpson Hurricane Wind Scale, that's some great background, that's how it started.

Oh. While I've got you, let's talk about Herbert Seymour Saffir, whose legacy in the hurricane world is so great that history has pretty much forgotten that in his youth, he was hired as a bellboy on this opulent wood-panel and silk-interioered cruise ship called *The Morro Castle*. Now, in September of 1934, Herbert would later recount, "I was sleeping at 4:00 AM when I suddenly was awakened by the bells." *The Morro Castle* that he was working on was engulfed in flames, partly because of bad wiring and partly because the wood-paneled interior was adhered with flammable glue, and partly because a lot of the crew didn't seem to give a rip about fire safety protocol. Also, the ship was sailing into a storm, which literally fanned the flames.

Now, half the lifeboats were in cinders, but 85 people aboard managed to get in them, and they were mostly crew members who were like, "Hurry, before we get swarmed with the dying people." And at this point in my research, I despised Herbert Seymour Saffir, even though this was when he

was young, and he later made the hurricane scale. But then I read further to learn that Herbert was not in a lifeboat during the shipwreck. He was trying to help passengers get to safety, and their only avenue was jumping five stories off the deck of the flaming *Morro Castle* into the water. Now, the life vests, they were difficult to wrangle, and some people's necks snapped as soon as they hit the water.

Now, in total, it's estimated that up to 137 people died in the shipwreck. But Herbert Seymour Saffir obviously survived because he created the hurricane scale later. He survived the fall, with this great gash to his head, and together with these two other passengers, an Italian couple from Detroit, they clung together like a clot in the sea, in the waves for seven hours, until the Coast Guard found and rescued them. And Herbert Seymour Saffir was so injured and weak, he almost accidentally drowned the woman he was clinging to the whole time. He kept choking out the words, "I'm sorry," which seems so polite, and then he was resuscitated at the shore. And he survived to become a civil engineer and a co-establisher of the Saffir-Simpson Hurricane Wind Scale.

Now, poor Simpson of this duo, he must have had, like, a super boring life. And then I remembered, well, he was a hurricane hunter. And really, no one's life is boring when you dig into it. And then I did some more digging, and I found out that Robert Simpson was married to the illustrious and highly lauded Joanne Simpson, who was the first woman to earn a PhD in meteorology. So, Dr. Joanne Simpson, fellow meteorologist, very much a BAMF in this space, she actually had three husbands in short succession. The first one, also a meteorologist. Another, a mathematician who worked on the physics of meteorology. And finally, Robert Simpson of the Saffir-Simpson Hurricane Wind Scale, who she met before she met her very first husband, and I suspect they were in love with each other the whole time. If they weren't, I'm sorry that I imagined it. But doctors Robert and Joanne Simpson stayed married until death did them part, and on their joint tombstone in Washington DC, right between their two names, engraved in stone, is the shape of a cyclone.

So yeah, the Saffir-Simpson Hurricane Wind Scale. Let's get to it. If you've got a tropical depression, all right, lowest on the scale. Then it gets upgraded to a tropical storm. Then Category 1 hurricane gets you the hurricane badge; 74 mile an hour winds, that's 119 kilometers. Category 2 starts at 96 miles an hour, or 154 kilometers; this is called "extremely dangerous with extensive damage." Category 3 starts at 111 miles an hour, or 178 kilometers; this is classified as "devastating damage." Category 4 hurricane, 130 miles an hour, or 209 kilometers; this is called "catastrophic." Category 5 hurricane, 157 miles per hour or higher, or 252 kilometers an hour; again, "catastrophic." So, tropical depression, tropical storm, and then hurricanes, Categories 1 through 5.

Kim: And then of course, there's the elusive Category 5, which, there's not much difference between a Category 4 and 5. If you actually go to the Saffir-Simpson Hurricane Wind Scale description, it is just labeled as "Catastrophic damage." So, Category 5 grabs attention, but it's not so much more damaging than a Category 4; Category 4 is pretty bad too.

Alie: Does it depend more on where they make landfall? When we're talking about catastrophic, I imagine we're doing that from a human lens, like, how many airports are we going to lose? How many homes are going to flood? Or does that not have to do with humans?

Kim: Originally speaking, we associated the maximum wind speed with the damage because we wanted people to be prepared for if you got hit with that maximum wind speed. But a hurricane has multiple hazards. We categorize them by their maximum wind speed, but their maximum wind speed is estimated over water, and water delivers less friction than land, so you almost never see those strongest wind speeds of its category actually measured over land because it's just somewhere in the storm. But also, a hurricane delivers heavy rain, it delivers storm surge along the

coast, and has the potential to produce tornadoes, because why not add insult to injury? [*"Please... Don't."*] And so, when we're talking about hurricane hazards, the challenge often comes from we are using this wind scale to categorize them.

Aside: Kim mentioned 2019's Tropical Storm Imelda, which, although it was not a hurricane, it did dump 41 inches of rain over southeast Texas, which is just about 85 miles from Houston, which just two years earlier, got smacked with the Category 4 Hurricane Harvey. Now, Harvey caused \$125 billion in damages and it took over 100 lives. Tropical Storm Imelda still caused seven fatalities and five billion in damages. So, you're not off the hook on a tropical storm technicality, Imelda.

Kim: So, just because it was a tropical storm doesn't mean it isn't capable of delivering outsized hazards that will bring all kinds of harm and infrastructure challenges to the people in its path. So, when it comes to category, it's one of those things where the categorization is useful, but to a point, because then all of us who work in this area need to add that nuance. That's where local meteorologists trusted by their communities come in so handy, they can personalize the information about what a storm could do.

Aside: Okay, but...

Alie: ValBeListening wanted to know: Are they considering adding more categories for hurricanes? CurtisDawg wanted to know whether or not there's going to be, like, more 4s and 5s than ever before. And then Brianne Roberts wanted to know: How likely are we to have a higher category, like, turning something up to 11? Sarahbeth actually is a disaster researcher, wrote in all caps: LET'S BANISH THE HURRICANE CATEGORIES! The scale isn't a great tool for capturing how "bad" it is because it only accounts for wind speed.

So, yes. Categories: Do you think they'll change? Do you think that they're a good way of warning the public? Do you think they get overblown in TV meteorology? Like, how do you feel about 'em? Lot of questions.

Matt: [*chuckles*] Yeah, no, no, no. But they're all hitting on a very, very important topic. We know hurricanes are complicated; they're more than just wind. It's wind, it's surge, it's rain, it's tornadoes, you know, it's all these different things. And the Saffir-Simpson Scale, which is the scale that we use to rate hurricanes 1 to 5, you know, based on their wind speed, is like you said, it's wind only. That's the only thing that goes into it. You could have a storm that's a little, itty-bitty hurricane, and you have 160-mile-an-hour winds, and it's going to impact a small patch of land. But it's still a Category 5 hurricane at that point, right? The impacts are going to be very limited to a very small area. And you know, if it's like far south Texas, where almost nobody lives between the Rio Grande Valley and Corpus Christi—I shouldn't say nobody... Very, very, very few people live between the Rio Grande Valley and Corpus Christi—you get a storm that goes in there, it's mostly pasture, that's what gets hit, and, you know, then it's no big deal.

Aside: Or less of a big deal. Unless you're a cow. And yeah, I went down a whole badger borough on what kind of pastures they got out there. And I found out there's one ranch called King Ranch that's bigger than the country of Luxembourg at nearly 1,300 square miles. A guy bought the land for \$300 in 1853, and I'm thinking by now they've probably even made a profit.

And if you think *Yellowstone* is a thrill ride, wait until you hear how one of the co-owners of the King Ranch was a guy named Captain Gideon K "Legs" Lewis, who was having an extramarital affair with one Annamarie Perlee. But Annamarie Perlee was married, and her husband, Dr. JT Yarrington, had killed a guy in San Antonio in a gunfight a few years earlier, but he got off. Now, when this already known killer, Dr. JT Yarrington, found out about his wife stepping out on him

with an East Texas cattle rancher who went by the name Legs, he, being already a killer, was not happy about this.

Now, Legs Lewis tried to visit his lover, Annamarie, to get back his love letters from her, when, according to obituaries, "JT cut Lewis in half with a double-barreled shotgun." Annamarie was granted a divorce from JT on account of the guy being a double murderer, and she lived a long life. Now, what happened to JT? Second killing? He got away with murder again, and he moved to San Francisco, he got a new lady, and they're buried together in Oakland, although both of their headstones are missing. Karma? Who knows?

But yeah, if a hurricane hits East Texas, some cows will likely be in danger because they've got quite a few cows out there.

Matt: So, when you're talking about categories, it's useful in the sense that it's nice to be able to scientifically categorize storms and understand trends in wind speed, things like that. From a public communication standpoint, frankly, it sucks. It distracts, I think, sometimes from the thing.

I mean, think about it, last year, we had Hurricane Beryl hit Houston. It was only a Category 1 storm. A lot of people will kind of shrug their shoulders at a category one and be like, "All right, that's a storm, but I don't need to go running for the hills because of that." And it comes in, it does billions of dollars in damage, does tremendous damage to the electrical infrastructure in a major metropolitan area. It was a Category 1 storm, but it was rapidly intensifying up to landfall. So, when you think about this, you have two different scenarios with hurricanes. The storm is either intensifying as it makes landfall, or the storm is weakening as it makes landfall. It's almost never perfectly stable.

So, if I tell you a Category 3 is coming, you're going to be like, "Oh crap, this is bad." If I tell you a Category 1 is coming, you're going to be like, "Oh, okay. Do I need to worry? I don't think so." And the reality is, it depends on the type of storm. So, that's where the category scale really becomes, frankly, useless at that point. And then you factor in rain. Hurricane Harvey was a big storm, did a lot of damage, it was a Category 4 hurricane when it made landfall in Texas, but the worst element of Harvey was the inland flooding that it produced and it was catastrophic. And the Saffir-Simpson Scale is going to tell you this thing's a tropical storm.

Aside: So, Harvey made landfall and then slowed, was downgraded to a tropical storm. But despite the tropical storm name, this cyclonic system continued to firehose Texas with the wettest North American cyclone on record. And again, over 100 people died.

Matt: But the problem is, people love the categories. People still want to know, "What is this? Is it a 1, 2, 3, 4, or 5?" And you're telling them, "Well, no. Maybe that's not the best way to do it." But "No, no. I still want my number." So, you're kind of fighting that battle too.

Alie: And what about in other countries, like where there are these typhoons? Do they have the same category numbers? Do they have different ways of measuring it or different warning systems?

Kim: So, each location affected by tropical cyclones will develop their own approach to notifying the public, that sort of thing, and categories do vary depending on what basin it's in, because every culture is different. So, *[laughs]* time to get pedantic. *["Let me have it."]* In the North Atlantic and East North Pacific, we use what's called the one-minute, 10-meter maximum sustained winds. So, that means over a minute, the wind speed, on average, was this value, 10 meters above the surface.

Aside: But as for the typhoons off the coast of Japan, China, Taiwan, the Philippines, again, these are the western North Pacific typhoons.

Kim: And then, if you go over to the Western North Pacific, where the Japan Meteorological Agency is in charge, they do a 10-minute average, and that means a lower number. So, they'll mention that they've got this typhoon, and it will be, say, the equivalent of 80 miles an hour. And we'll be like, "Pffff" ... Okay, maybe not that bad. *[laughs]* But we might dismiss it as, "Oh, that's just a Category 1." I don't dismiss Category 1s, but it is the lowest on the hurricane scale. But because they're using that 10-minute average, if you did a one-minute, it would actually be a higher value, which is real fun if you're trying to do a global assessment of tropical cyclones and trying to estimate them all the same way. Well, we don't. Depending on what basin you're in, it's one-minute, two-minute, three-minute, or 10-minute.

Aside: And where precisely does it become a hurricane versus a typhoon? Well, they decided it was once it crosses the human-made boundary called the International Date Line. Now, according to NOAA's National Hurricane Center, the majority of tropical cyclones occur in the Northern Hemisphere, and there are more typhoons in the Pacific than there are Atlantic hurricanes.

Also, according to the 2022 study, "A review of ocean-atmosphere interactions during tropical cyclones in the north Indian Ocean," the north Indian Ocean accounts for only 6% of the global tropical cyclones annually, but it accounts for more than 80% of the global fatalities from cyclones, mostly due to coastal flooding. In Part 2, we'll talk a little bit more about how economics factor into hurricane science and safety.

So, there are cyclones below the equator, but what about right on the waistband of the world?

Alie: Well, several people wanted to know about the equator. Nicky G, Joshua.YYZ, Lucas, Chris Lipford, Evelyn Uyemura, Adz the Erminologist, Wendy Miller, Pierre Gregoire, Dave Brewer, and LightBrownPillow wanted to know, in Nicky's words: Why don't hurricanes cross the equator? Is there just, like, a stoplight that never turns? What's going on there?

Matt: So, the Earth rotates, right? As that happens, you've got different forces acting on different sides, right? Different sides of the equator. So, hurricanes always instinctively want to go toward the pole. You can't think of a hurricane necessarily as a living thing, but you know, that's what it's attempting to do. It's attempting to transfer heat from the tropics to the poles. And that's how the Earth kind of stays in balance, right? It's part of living on Earth. We always think of hurricanes as bad, but they're actually part of a well-functioning, normal Earth system. You'll see some storms get pretty far south sometimes. Like, you can get storms that are below 10 degrees latitude in either hemisphere, but they either can't survive because there's not enough spin and not enough force to keep them rotating, or they end up just going toward the pole. And that's that. It becomes impossible for them to survive below a certain latitude.

Aside: Okay. But is that forever?

Alie: Several people asked whether or not different parts of the world will have hurricanes in the future. Kelly Shaver wanted to know: Are we going to start getting hurricanes in California as the climate keeps changing? Madeliene Fox, first-time question-asker, asked: As climate change continues to shift our weather patterns, do we expect more hurricanes to move more inland? Clemens V wanted to know: Is Europe going to face hurricanes in the future under more progressed climate change? How far north or south? Lauren Cooper asked: So... climate change. How fucked are we?

Aside: We're going to get deeper into that question in a bit, but...

Matt: Yeah, I think we're going to see some changes, and I think we've already seen some evidence of that. Just kind of in recent years, we've seen storms show up that are in, like, the far northeast Atlantic. But we actually had a technical tropical storm, I think, hit Portugal a couple of years ago. I think the California question is interesting because, in California, they have been hit by tropical

systems before. And I think as the West Coast warms a little bit, you know, storms are still going to weaken as they come north out of the Eastern Pacific, but maybe now they just kind of linger a little bit longer.

You know, so when we think about climate change, we have to think about it as a multiplier on top of. I always say, like, the cake's baked, right? You're going to have cake. Now, you're just adding icing to that cake; you're increasing the number of calories. And that's what we get with climate change. So, things that used to happen, you know, now we're getting more of it. There's not always going to be a glaring, flashing red light that, "This is climate change." It's going to be kind of quietly seeping into everything weather-related, including hurricanes.

Alie: Now, what about busy time of year? Right now, we are in North America. We just had catastrophic flooding in Texas. I live in LA; we were evacuated for the fires. 2025 is already shaping up to be very big meteorologically. So, it very much feels like a when and not an if, knowing that you're just going to get busier and busier. But what is the hurricane season like?

Kim: So, for the North Atlantic, that's June 1st to November 30th. That said, in the North Atlantic, the peak of the season, when you're most likely to see at least one storm active, tends to be in the August to October timeframe. [*"Like, right now?" "That's right. Right now."*] So, even though the season officially begins June 1st, not seeing a lot of activity in June and even July is, on average, normal. Is anything about this year normal? No. But you know, that's when we tend to see tropical cyclones because that's when seed disturbances, like what I talked about coming off of Africa, tend to be emerging, and thus, offering the opportunity to become a tropical cyclone. Because a tropical cyclone doesn't just pop out of thin air.

Aside: Well, it is a low-pressure system, so I guess it does pop out of thin air, but then it rises and condenses to form pretty thick, wet air, I guess.

Kim: Something needs to be there to kind of galvanize it. But when it comes to looking at the upcoming season, we watch for things like, what is the activity over Africa like? How many seed disturbances are coming over the ocean? Are the waters warm, or warmer than normal?

Aside: And then there are factors like wind shear, which can lift or lower the storm system vertically, or it can shift the tilt horizontally. The greater the wind shear of that column of air, the less likely it is to make a cyclonic storm because it makes that swirling bit kind of wonky and the storm gets, like, discombobulated.

Alie: Basic question: Why a circle? And what's going on in the eye of a hurricane? Why is it so calm? What's happening?

Kim: [*laughs*] So, there's some pretty complicated physics at play, and two main contributors to the fact that an eye is present in the first place.

So, a storm needs to be a certain strength to even have an eye. You can start to see an eye wall—which is the really deep thunderstorms that surround the eye—you can start to see that form in radar when it's, like, a strong tropical storm. So, it doesn't have to be at the hurricane wind speeds necessarily to start forming that. But to get the clear eye, it needs to be somewhat strong because you're dealing with a lot of balances of forces. You've got winds moving air in, but then you also have this sinking area because it's warm and clear, and most of the air at the top of a storm gets evacuated out and away from the center. But some of it also comes back in, and when air sinks, it gets compressed.

Alie: Oh! And then, is the eye of that storm calm? Again, never been in a hurricane, but I picture it being absolute hell on Earth, chaos, terrible, terrible, and then [*sings like an angel*] "Ahhhh," and this

impending feeling of doom that it's coming back. Is that kind of how it is, or is that just from the movies?

Kim: Honestly, what you just described is pretty accurate, but there's a peer-reviewed paper that talks about the ill-fated flight into Hurricane Hugo. Usually, scientific writing is dry. It's written passively; it's just trying to state the facts and argue why their conclusions make sense. There's a paragraph in this paper that, for scientific writing, reads like poetry, because it's talking about the experience of losing an engine while in a hurricane. They talk about how far they fell, how close they got to the surface, the white caps they could see on the ocean below because they were in the eye. So, they kind of limped along, slowly gaining altitude back up on their remaining engine until another plane could come in and guide them out.

Alie: Oh my god, goosebumps.

Kim: And I'm like, why isn't this a movie?

Aside: And while you are sketching out a screenplay, or you're finding your umbrella, we're going to take a quick break. After that break, you will hear all kinds of info about hurricane names, masculine, feminine, advice on preparing a storm, what's going on exactly with federal funding, and if you can stop a hurricane in its tracks, yet. Some of that continues into Part 2 for next week. You *definitely* want to hear that.

But let's hear from sponsors of the show who make it possible for us to donate to a cause of the ologist's choosing. And this week, we'll go with Matt's. He selected the Community Foundation of the Texas Hill Country's Kerr County Flood Relief Fund, which supports urgent relief and long-term rebuilding after the devastating floods of the 4th of July, 2025. So, a donation will be made in his name. Thank you to sponsors of the show for making that donation possible.

[Ad Break]

Okay, we sprinkled in your Patreon questions throughout the episode, but let's stir up some more answers.

Alie: You know, speaking of a lot of things in pop culture and what we hear about hurricanes, there's this adage, I don't know if it's flimflam, but if it has a female name, people don't pay attention to it, and then more people get killed because they don't hide from it. I don't know if this is flimflam, I don't know if this came from, like, a Malcolm Gladwell book that everyone forgot about. What is the deal with this?

Kim: Ahhh, so there's a story here. Are you familiar with the *Clue* meme "Flames on the side of my face"? [*"So... much. It, it, it— F- F- Flames. Flames, flames on the side of my face... breathing."*]

Alie: "F- F- Flames on the... Heaving, burning." The most. Yeah, it's in my DNA. [*laughs*]

Kim: Yeah, when this paper first came out in a journal I respected until then, when I was double-checking some things ahead of our conversation, I came across that again and just had that visceral reaction that mimicked what I saw a decade ago when it did first come out, claiming that somehow female-named storms are deadlier than male-named storms. What I was gratified to see in that quick search is that two—not one, *two*—independent rebuttals came out in the same journal saying, "No, this is not true and the statistics show it. Why does this even exist?" So, you know, I still ask that question. But yeah, it definitely got sensationalized. It also got scientifically debunked. So, no. There is no statistical correlation between the perceived gender name of a tropical cyclone and its impacts.

Aside: So, the article in question was the 2014 paper titled, “Female hurricanes are deadlier than male hurricanes,” in the journal *PNAS*, which Kim told me is, no joke, sometimes called [phonetic] “Pee-nas.” Unfortunate. But this paper asserts that “US hurricanes used to be given only female names, a practice that meteorologists of a different era considered appropriate due to such characteristics of hurricanes as unpredictability.” Also, male scientists named them after girlfriends and wives because... pee-nas.

Now, the guy who started the lady naming was one, a 19th-century British meteorologist. His name was Clement Wragge, who is described as having “an iron constitution, a mop of flaming red hair and an explosive temper to match.” Now, his villain origin story is that he was very sadly orphaned at age 5 when his dad fell off a horse. But it also sounds like he was left with a hefty trust fund, and rage issues, and a lot of enemies. So, he started naming storms by Greek letters of the alphabet and then by Greek gods and goddesses, and then he moved on to naming the worst storms after those enemies, many of whom were politicians who would not fund his research, which.... It’s interesting. It could be interesting. Now, Clement Wragge then moved on to naming hurricanes after women. He thought that giving them exotic names could maybe influence baby names of the time, and they did. You know, after hearing these names of storms, there were more baby girls, Eline, and Luita and Leala.

But what’s in a name? Drama, lots of drama. Just ask Clement Lindley Wragge himself, who was actually born William Lindley. Now, Clement’s wife’s name was Leonora, and they had a daughter, also Leonora, later renamed her Emma. Clement’s dad was named Clement, and so was his cousin. And Clement’s first male child was also called Clement. When Clement and Leonora, not the kids, but the parents, split up, Clement found another lady named Louisa, who everyone called Edris.

So, my point is that Clement Wragge had name trauma, and because of that, we have gender reveals in our weather forecasts, and it took so many decades and women who bravely wore pants in the 1970s to finally say, “Can we get some man names in the mix, at least?” I bet Dr. Joanna Simpson was one of them.

Alie: Which, by the way, I feel like gendered names aren’t even really a thing that much anymore. Do you know what I mean? Like... Why? It’s actually very exciting, too. I get to talk to a non-binary meteorologist, like, when does this get to happen? And I’m like, “Who’s going to have the hottest take ever? Actually, you.”

Kim: *[laughs]* So, if you look back at the history of naming, it started in the 1940s. So, the story is that storms got named after the wives, sisters, or girlfriends of guys in the Navy. And so, that started the tradition of female names. *[Alie moans softly]* I think it was 1953, if I recall correctly, where the National Weather Service tried to do something a little bit more consistent using phonetic letter type names.

Aside: Like Alfa, Bravo, Charlie. But in the middle of that naming experiment, the International Phonetic Alphabet had to be changed because it originally had N and V as Nectar and Victor, but they sounded too close, so they ended up changing the alphabet. Anyway, a bunch of dudes in ties probably gathered around a big conference room, smoking over low balls of whiskey and just said, “Fuck it, gents.”

Kim: So, they went back to female names, and then they started being more consistent and alternating between male and female.

Alie: Are they looking to change that system at all? It’s like, not cute, you know? I don’t know, maybe it’s cute, but I’m like... This idea that it’s like, “We are having a baby!” *[Kim laughs]* It’s going to kill a lot of people, and climate change is getting worse.” Do you know what I mean?

Kim: Just as an aside. Something that happens because we give them gendered names is we will call storms he or she, and I'm like, "It! A storm does not have a gender."

Alie: *[laughs]* That's so funny that it's, like... Especially with current discourse about what is gendered and what's not. They're like, "That storm's definitely a boy." I'm like, "Is it? Okay." But I mean, do they have a Google Doc somewhere of baby names for it?

Kim: There's actually a public document on the National Hurricane Center webpage of their six rotating lists. So, as far as where the names come from in the first place, there are committees at the World Meteorological Organization, or WMO, where people get together to review, one, whether names should be retired when they have very high impacts, or when the name itself is fraught with meaning that no longer needs to be used as a name. Two names that got retired in that regard were Alfred and Isis.

Alie: Oh. Oh.

Kim: Yeah. So, those aren't used anymore. But those committees are formed of representatives of the countries in the regions affected by those storms. But they alternate. So, you may notice, in the North Atlantic, if the A name is male, it'll be female in the Eastern Pacific. So, they mirror each other that way.

Aside: And remember that, yes, the eastern North Pacific coast off Mexico and Central America and the Baja Peninsula, and yeah, California can get hurricanes, and tempestologists are always trying to get people to remember that underdog of the eastern North Pacific hurricane basin. We care about you.

Alie: But you know what? Given the way the world is, I feel like it's just a matter of time before it becomes like, the Pepsi Category 5 or, like, the Crypto Tropical Depression. You know what I mean? Or it's going to be like, Hurricane McHurricane Face if we take it to a vote. *[Kim squeals with laughter]* Maybe there are worse options. It's like you don't know when you have 'til it's gone.

Aside: What is Matt's naming forecast?

Alie: In terms of naming, Meghan Ratcliffe wanted to know if you know why they're named after people names? Why not animals or fruits or literally anything else? Has there ever been a move to just name them something else? Like, you know how Apple iOS started out as like "Tiger, Panther," and then they were like, "We ran out of cats. So, now it's Cupertino and Napa," and you're like, okay.

Matt: Let's keep going. We're going to be down to zip codes soon. *[laughs]*

Alie: *[laughs]* Seriously.

Matt: I've always joked it would be kind of fun to, like, have storms that are like sponsored by corporate entities. You know, so like, "This storm brought to you by Lysol."

Alie: Oh my god. *[both laugh]* I talked to Kim about that, that there's going to be like a Pepsi...

Matt: You're right! *[laughs]*

Alie: Oh my god, that's so real. Oh my god, Yeah, what's going to happen?

Matt: Who the hell knows? *[“Not me.”]* What they do now is, you know, it's actually really a formal deal, is that the World Meteorological Organization, which is essentially like the UN for weather, gets together in Geneva annually. They go back through all the tropical systems that happened the previous year, and the ones that were really awful, they retire the names of, and they replace them with a new name. What happens is, like, every country that's in a certain basin that uses that name

list gets to submit a name that kind of reflects their ethnicity and their country, and submits a name, and then they pick one. But it's a real formal, kind of a weird process. You know, I'm a Matthew. Matthew was...

Alie: I was going to say, how...

Matt: You know, I'm already retired.

Alie: 2016, right?

Matt: Yeah!

Alie: How did you feel about that? How did you feel in 2016, when they're like, "Aaaand it's a Matthew," Were you like, "Yes!"

Matt: *[laughs]* No. I've never wanted it to be like a destructive storm. I was like, give me a Matthew that's a Cat 5 in the middle of the ocean, that's like, the coolest storm you've ever had to look at, and hurts no one. So yeah, it was weird. It was kind of like, funny that, "Okay, well, here we are, Matthew's on the list. And it's gone. Retired. All right, cool." Well, it was a fun run.

Aside: So, NOAA is predicting a banger of a hurricane season — above average and with possibly three to five major hurricanes. Now, some predetermined storm names you can look forward to in the new cycle are Dexter, Ferdinand, and yes, Karen, which I imagine will be dumping inches and inches of warm Chardonnay and uprooting Yelp! pages with bad reviews.

But before we can name these hurricanes, we have to track them, and we've got to model them.

Alie: And how does that modeling happen? Have you seen that modeling change just even in the last few years? I picture you putting a bunch of ones and zeros in a computer and then pressing a button, it goes "beep, beep, boop, beep, beep, boop, boop," *[Kim laughs]* and then it tells you the address that it's going to make landfall. But how does it actually work?

Kim: The workflow of what's going on is very complicated and requires a lot of infrastructure, all actually talking to each other; so servers not going down, satellites actually transmitting, that sort of thing. But initially, the workflow is we're getting all these observations from weather balloons, from sometimes radar (radar is a little complicated in that regard, just because it's only over land), but also from satellites, because satellites get that bird's eye—well, bird in space eye—of what's going on in the atmosphere. And through a process called data assimilation, we bring all that information together to then give the computer model, which a whole bunch of talented scientists and engineers have worked on to make sure it's as good as it can be for now, and then continue tweaking over time. That model takes those initial conditions, and if a hurricane is currently active, or a tropical cyclone is currently active, and planes have been flying through or around it to take measurements, those will be brought into those initial conditions to help the model better capture the current state of the atmosphere so that we're starting from a more accurate point to predict what it'll do next.

And then, the computer model is running a whole bunch of physics equations and making certain assumptions depending on how the model is set up, because we can't simulate at every single point in existence, it would never be done in time. So, then the computer model will produce these forecasts of what the atmosphere could be like in six hours, 12 hours, so on and so forth. Those outputs help us identify things like, do we think the storm has certain potential to get strong really fast, something called rapid intensification? What might it do size-wise? Is it getting bigger, smaller, that sort of thing? And then where is it going to go? But if you're trying to visualize what could be making a storm move a certain way, it's the flow of the atmosphere around it.

Alie: That makes sense.

Kim: And the computer models are helping us capture that aspect as well as what the storm itself is doing.

Aside: And yeah, in Part 2, we're going to chat about funding cuts and staffing layoffs. Literally, as I put this episode out, those votes are still on the table. Every hour, it seems like there's new information. Also, we're going to have a lot of information on how to prepare, what to do in a hurricane, and how to know when it's time to get up and go.

Matt: Yeah, the weather balloon issues a big wild card in all of this because if we do lose some of this upper air data consistently, then it becomes a problem; it does start to infect the modeling, modeling gets worse, things like that. So, you know, hopefully, as we go through the rest of hurricane season, you know, we're not missing too much of that data. Some of the weather balloons have started to be launched again. There's still some that aren't, but I imagine if you've got a Category 3 hurricane sitting in the Caribbean, they're going to launch the balloons, they'll find a way to do it. At least that's in Matt's world, that's what makes sense... And sometimes Matt's world is kind of a fantasy, sometimes. *[laughs]*

But the reality is that yes, weather modeling has gotten way better. We have invested a ton of time, research, energy, funding, et cetera, since Hurricane Katrina to better predict hurricanes, but it remains a very broad, bipartisan issue that needs to be continued to be addressed. 2024 was the best year on record for the National Hurricane Center in terms of their forecasting; they knocked it out of the park. They did a phenomenal job. Perfect? No. But way better than it has been. In fact, like, you look today, your five-day forecast today is almost as good as, like, a one- or two-day forecast was 30, 40 years ago.

Alie: Oh, wow.

Matt: I mean, think about that for a second, think about how much how many decisions get made in those few days before a hurricane by public officials, by companies, by anyone, you know, people that live on the coast, and now you have the confidence that you can start making moves four or five days in advance that maybe 30 years ago, you could only do one or two days in advance. So, that's why we say all these improvements and forecasts have huge economic and sociological benefits to society, because of that.

Well, now with the advent of machine learning and AI, we're starting to see that, okay, well, maybe we don't need to model the physics of the atmosphere. Maybe we can say, "Hey, here's 50 years of weather data," what we call reanalysis data. It's basically all the past weather that's occurred over the last 50 years. "Here it is. Take a look at this. Here's what's happening right now. Based on that history, tell me what's going to happen over the next 10, 15 days, whatever." And these AI models will be like, "Okay, sure. Here you go. Press the button. Go make yourself a sandwich. Come back, model is done." *[Alie laughs]* And you can get an answer. But I mean, I'll be honest, like, last hurricane season, the AI models did a pretty good job on forecast tracks with some storms.

Alie: Oh wow, okay.

Matt: I was floored. And it was kind of a wake-up call. We're still working with the physics-based models, we're still working to improve them, because we're not going to get rid of them. We're working into a world now where we're going to end up doing, like, a hybrid approach where you're trying to marry the two outputs and come up with an answer that's, you know, better than anything we've had before, and I'm super excited. I really think that in the next five to 10 years, we're going to have so much more interesting and useful, and valuable data. The caveat, of course, is: As long as

we continue to fund it, we will get better at this. We'll never be perfect, but we'll get better, which is great. That's what we need.

Alie: Well, you know, you mentioned Hurricane Katrina and the anniversary too, that 20th anniversary is coming up this year. Was there something about Katrina that made it so devastating? Was it the category? Was it where it fell, with New Orleans being below sea level? If hurricane and cyclone, and typhoon risk is twofold, partly that we're getting more of them because of warmer weather and partly our interface with the coast and very vulnerable areas, if that's just increasing... Like, is it where you live, or is it how hard it hits?

Kim: It's a combination of both. So, the infrastructure we build in places at risk of hurricane impacts, ideally, would be built with those potential impacts in mind. They often either aren't, or they aren't consistently. So, you know, if it's been a while since a hurricane has hit, people might be like, "Oh, I don't need to invest all this extra money or time into making sure this house is exactly up to these standards." And so, it's like, who's going to pay the bill? And if everyone fights over who pays the bill, well, maybe it just doesn't get paid. So, the infrastructure element is a component of it, and us building in these places that could be impacted, but the scope of the storm also matters.

So earlier, we were talking about the Saffir-Simpson Hurricane Wind Scale, and that's how we categorize storms. Katrina was devastating as a Category 3. It did peak as a Category 5, but it was weakening on approach to land, and seeing the category numbers tick down does have a psychological impact to anyone who is associating those numbers with impacts.

Another thing is, structurally, weakening storms do different things. So, oftentimes you'll see increasing compactness of, like, the core of the hurricane as it's getting stronger. Well, when it starts to weaken, that core can expand. And sure, the peak wind speed is going down, but you're making the area over which there are winds, bigger. So, when a storm is growing in size, the part of the ocean that that wind is interacting with gets bigger. So, you're driving more water toward a vulnerable coastline. If you look at animations that follow Katrina's lifetime, the eye was compact, and then it got big. And yes, the maximum wind speed went down, but the storm grew. So, it put more people in the path of strong winds.

Something that's hard to think about but is important to think about is a tropical storm is pretty strong. So, if you get tropical storm force winds, that's sticking your arm out as you're going 40 miles an hour. If you're going 70 miles an hour, that's even stronger if you stick your arm out the window; that's a tropical storm. That's not even a hurricane yet.

Alie: I never even realized that you could stick your arm out the window of a car, and that simulates how fast the winds are going. That never occurred to me.

Kim: And they're gusty, so you'll get, like, the variance. So, you know, as they get stronger and stronger, it'll feel more consistently brutal. But you'll also kind of get these pulses where it's, like, a strong gust comes through, and then it calms down a little bit, but it's still windy. So, even if it's a persistent tropical storm force winds, sitting in 40-mile-an-hour winds for an hour sounds awful.

Alie: Yeah. Is that what a sustained wind, when they say sustained wind? Okay.

Kim: Yeah. So, that's a good way to contextualize. Like for a minute, at least, you're getting hit with 40-mile-an-hour winds. For a minute, at least, you're getting hit with 100-mile-an-hour winds.

Alie: Oof! Oof!

So, ask delightful people disastrous questions. And thank you both so much, Matt and Kim, for not being an indie electronic duo, but for being two of the finest tempestologists a person could chat with. So much so that we'll be back next week to cover how to prepare for a hurricane, or really any disaster that comes your way, what happens behind the scenes when a president marks up a hurricane map on TV, where should you retire, and what is up with FEMA and NOAA, and the National Weather Service? All that plus some more weird history.

So, that'll be next week. Until then, you can follow Matt and Kim at their social media handles. On Bluesky, Matt is @MattLanza, Kim is @DrKimWood, which are linked in the show notes, or on our website at AlieWard.com/Ologies/Tempestology. We are @Ologies on Bluesky and Instagram. I'm @AlieWard on both. *Smologies* are those shorter, classroom-safe versions of *Ologies* classics available wherever you get podcasts. You can sign up at Patreon.com/Ologies to send in your questions ahead of recording. Erin Talbert admin's the *Ologies* Podcast Facebook group. Aveline Malek makes our professional transcripts. Kelly R Dwyer does the website. Noel Dilworth tracks our unpredictable schedule. Susan Hale oversees it all as managing director. And pilots of our always-exciting and shaky flight to hit publish are Jake Chaffee and Captain Mercedes Maitland of Maitland Audio. Nick Thorburn wrote the theme music.

And if you stick around till the end of this episode, I tell you a secret. This week, it's that... So, Jake spells his last name, C-H-A-F-F-E-E. But the first time I said it on here, I said [ph.] "Cha-fee." And for some reason, in the year-plus he has worked on the show—we talk, like, every day, it's been 50 or 60 episodes at least—I have to write his name in my notes with a Y in there because I'm so worried that I'll say "Cha-fee" and not "Chay-fee." And I could say these credits in my sleep. And I know he sees my notes in the Google Doc, and I spell his name wrong every time, but on purpose, for phonetic reasons. Now I feel like it would be weird to spell it right, like it would throw me off.

Also, speaking on the notes for this episode, the Google Doc, the working transcript for this episode, which is all color-coded with sound effects and asides and all that, is 63 pages long. So yeah, you're getting a two-parter, my babies. We will see you next Tuesday, and I mean that in a nice way. Okay, berbye!

Links to things we discussed:

[How Eric Berger and Matt Lanza Became Houston's Most Trusted Meteorologists](#)

[Hurricane Science Was Great While It Lasted - The Atlantic](#)

[FEMA: Mitigation Saves: Federal Mitigation Grants Save \\$6 per \\$1 Spent](#)

[What is the difference between a hurricane and a typhoon?](#)

[Websites hosting major US climate reports taken down](#)

[The Hurricane Hunters That Never Returned](#)

[Full video of Hurricane Hunters flying through tough weather](#)

[High vs. Low-Pressure Systems Explained](#)

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[King Ranch: Secret History](#)

[Who was Jacob Tier Yarrington?](#)

[Tropical Cyclone Naming History and Retired Names](#)

[TWISTER Clip - "Cows" \(1996\) Bill Paxton + Helen Hunt](#)

[June 2025: Fired, rehired and fired again, some NOAA employees get letters demanding money](#)

[July 2025: Cuts to NOAA increase the risk of deadly weather tragedies](#)

[July 2025: Congressional committees push back on Trump administration's proposed NOAA budget cuts](#)

[Herbert Seymour Saffir's *Morro Castle* Survival](#)

[Severe Weather Topics - Wind Shear](#)

[NOAA's National Weather Service - Glossary - shear](#)

[25th Anniversary of a 'hairy hop' into Hurricane Hugo - NOAA's Atlantic Oceanographic and Meteorological Laboratory](#)

[The Wildest Ride on a Hurricane Hunter Aircraft - Eos](#)

[Structure of the Eye and Eyewall of Hurricane Hugo \(1989\)](#)

[Clement Lindley Wragge: The Life of a Pioneer Meteorologist](#)

['Inclement' Wragge : pioneer weather forecaster](#)

[Deaths Related To Hurricane Rita And Mass Evacuation"](#)

[Encyclopedia of Hurricanes, Typhoons, and Cyclones, New Edition](#)

[A review of ocean-atmosphere interactions during tropical cyclones in the north Indian Ocean](#)

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